Grade 10 Global Economics — Term 1 Review Outline

2026–2027

1 Introduction

First, identify what the decision maker can choose and what the world will reveal later. Next, organize the possible outcomes in a payoff table. Once the payoffs are clear, compare decisions under uncertainty, test how a recommendation changes, and apply criteria in sequence when the problem calls for several rounds of filtering.

Review routine. Identify the choice the decision maker controls. Organize outcomes by alternative and state, with a payoff measure and unit. Check the available probabilities and constraints. Apply the appropriate rule, show the values you compared, and justify the recommendation in context. Name all tied alternatives when applicable.

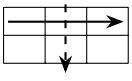
Scan the summary to recognize the task and find the method. Follow a criterion link to check your setup, intermediate calculations, and conclusion against a worked example.

2 Summarized Outline

	Cue in the question	Method	Write or draw
C1	Describe the decision.	Sort choices and states.	Who chooses, when, and what each payoff measures.
C2	Build a contribution table.	Add rates, scale, add fixed sum.	Formula, substitution, table.
C3	Build a profit table.	Revenue less both costs.	Revenue, costs, profit by cell.
C4	Optimistic or conservative?	Compare row maxima or minima.	Row statistics and winners.
C5	Limit missed profit.	Column bests, then row regrets.	Benchmarks, table, lowest max.
C6	Use state probabilities.	Find the weighted sum per row.	Probability check, terms, EMV.
C7	When does the choice change?	Find feasible bounds and EMV tie.	Equations, graph, intervals.
C8	Compare rules or change a payoff.	Update, screen, and rank.	Winners, values, tie or threshold.
C9	Filter options in stages.	Rank the survivors at each step.	Filtering table and tree.
C10	Justify a rule in context.	Match information and priorities.	Rationale and webpage checks.

Four rules for payoffs where more is better

Rule	Prob.?	Statistic	Choose	Priority	Limitation
Maximax	No	Row maximum	Highest	Greatest upside	Ignores downside.
Maximin	No	Row minimum	Highest	Best worst case	Can pass up large gains.
Minimax Regret	No	Row maximum regret	Lowest	Least worst missed profit	Depends on the comparison set.
EMV/Bayes	Yes	Row weighted sum	Highest	Highest expected payoff	Depends on probability estimates.



Across a row (solid) gives one alternative's maximum, minimum, EMV, or maximum regret. **Down a state column** (dashed) gives the best payoff to use as that state's regret benchmark.

3 Step-by-Step Outline

In the examples, gray shading traces a payoff from its definition or calculation into a table or comparison. Purple shading follows a calculated regret into the regret table. Blue shading links state probabilities to their weights in EMV. **Bold deciding values** mark the statistic or threshold that determines the recommendation. The graph emphasizes the tie, and the tree emphasizes the final selection. Always carry the original monetary unit through the calculation.

C1 — Identify the decision elements

When to use it. The question asks you to characterize a choice under uncertainty.

1. Name the decision maker, the economic purpose, and the alternatives they control.
2. List the states of nature in order and say when the actual state is revealed.
3. Explain how the state changes the outcome and name the payoff measure.

Craft Workshop. The studio chooses a workshop to earn a net contribution.

Element	Description	
Alternatives	A_1**:** Pottery workshop	A_2**:** Weaving workshop
States of nature	S_1**:** Strong tourist demand	S_2**:** Weak tourist demand
Uncertain event	The studio cannot control tourist demand. It learns the demand state after choosing.	
Consequences	Demand changes attendance and the extra contribution per participant. Each payoff is the total net contribution in USD for one workshop and one demand state.	

For Pottery under strong demand, $12 + 6 = 18$ USD per participant before sponsorship. The studio controls the workshop choice. Demand determines which of that workshop's payoffs it earns.

C2 — Build a payoff table from contributions

When to use it. The givens include per-participant contributions, attendance, and a fixed amount.

1. Put alternatives A_i in rows and states S_j in columns.
2. Add the workshop rate v_i and the demand addition d_j, both in USD per participant.
3. For each cell, multiply by attendance q_{ij} and add sponsorship f_i once, $P_{ay}(A_i, S_j) = q_{ij}(v_i + d_j) + f_i$.

Craft Workshop, continued. Pottery and Weaving contribute $12, 15$ USD per participant and receive sponsorships of $80, 50$ USD, respectively. Strong and weak demand add $6, 2$ USD per participant.

Net contribution (USD)

Symbolic calculations.

	Strong (S_1)	Weak (S_2)
A_1	$q_{11}(v_1+d_1)+f_1$ $= P_{ay}(A_1, S_1)$	$q_{12}(v_1+d_2)+f_1$ $= P_{ay}(A_1, S_2)$
A_2	$q_{21}(v_2+d_1)+f_2$ $= P_{ay}(A_2, S_1)$	$q_{22}(v_2+d_2)+f_2$ $= P_{ay}(A_2, S_2)$

Numerical calculations.

	Strong (S_1)	Weak (S_2)
A_1	$30(12+6)+80$ $= 620$	$18(12+2)+80$ $=332$
A_2	$24(15+6)+50$ $=554$	$16(15+2)+50$ $=322$

Final payoff table (net contribution in USD).

Alternative	Strong (S_1)	Weak (S_2)
Pottery workshop (A_1)	620	332
Weaving workshop (A_2)	554	322

Decision. The Pottery workshop earns the higher contribution under both strong and weak demand.

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C3 — Build a profit table from revenue and costs

When to use it. You have sales quantities, revenue per sale, variable costs, and fixed costs.

1. Multiply places sold q_{ij} by base revenue b_i plus the per-sale surcharge s_j.
2. Multiply q_{ij} by the variable cost per place c_i.
3. Subtract total variable cost and the fixed seasonal cost F_j once. Enter the profit in its row and column.

Guided Cycling Tours

Town and Countryside have base revenues $32, 45$ and variable costs $18, 26$, in USD per place. Strong and weak demand add $8, 3$ USD per place and incur fixed seasonal costs of $500, 650$ USD.

Trace Town under strong demand (USD)

Symbolic calculations.		Numerical calculations.
Component	**Expression**	**Substitution for 90 places**
Revenue	$R_{11} = q_{11}(b_1 + s_1)$	$90(32+8) = 3600$
Variable cost	$VC_{11} = q_{11}c_1$	$90(18) = 1620$
Profit	$P(A_1, S_1) = R_{11} - VC_{11} - F_1$	$3600 - 1620 - 500 =$ 1480

Route	Demand	Places	Revenue	Variable cost	Fixed cost	Profit (USD)
Town (A_1)	Strong (S_1)	90	3600	1620	500	1480
	Weak (S_2)	60	2100	1080	650	370
Countryside (A_2)	Strong (S_1)	70	3710	1820	500	1390
	Weak (S_2)	40	1920	1040	650	230

Final payoff table (seasonal profit in USD).

Alternative	Strong (S_1)	Weak (S_2)
Town route (A_1)	1480	370
Countryside route (A_2)	1390	230

Decision. The town route has higher seasonal profit in both demand states.

C4 — Compare Maximax and Maximin

When to use it. The question asks for an optimistic choice or the best protection against a poor outcome.

1. For Maximax, find the maximum payoff M_i in each row and choose the highest M_i.
2. For Maximin, find the minimum payoff m_i in each row and choose the highest m_i.

Sportswear Stock

A retailer chooses a stock plan with no demand probabilities. Profits are in thousand USD.

Alternative	High	Medium	Low	Row max M_i	Row min m_i
Bulk order (A_1)	120	48	−30	**120**	−30
Regular order (A_2)	84	57	16	84	16
Small order (A_3)	55	43	29	55	**29**
Pre-orders only (A_4)	24	24	24	24	24

Decision. Maximax selects **Bulk order**, with an upside of **120** and a possible loss of 30. Maximin selects **Small order**. Its minimum, $\min\{55, 43,$ 29 $\} =$ **29**, gives the best protection, with an upside of 55.

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C5 — Minimize maximum regret

When to use it. The decision maker wants to limit the largest profit they could miss.

1. Compare down each state column to find its best payoff b_j.
2. Subtract each payoff from its column benchmark, $r_{ij} = b_j - P(A_i, S_j)$.
3. Find the largest regret in each row and choose the smallest of those row maxima.

Flower Stall

A florist chooses a purchasing plan before holiday demand is known. Regret is the extra profit the florist could have earned by choosing the best plan for the state that occurred. Both tables use thousand USD.

Payoffs

Alternative	High	Medium	Low
Large purchase (A_1)	78	32	−12
Medium purchase (A_2)	60	48	18
Small purchase (A_3)	40	38	30
State best b_j	78	48	30

Regrets

Plan	High	Medium	Low	Row max
A_1	0	16	42	42
A_2	18	0	12	**18**
A_3	38	10	0	38

For A_2, the regrets are $78 - 60 = 18$, $48 - 48 = 0$, and $30 - 18 = 12$. For A_1 under low demand, subtracting a negative payoff gives $30 - (-12) = 42$. The smallest row maximum is $\min\{42, 18, 38\} = \mathbf{18}$.
Decision. Choose **Medium purchase** (A_2). Its largest possible missed profit is **18** thousand USD.

C6 — Calculate EMV and apply Bayes

When to use it. The question supplies state probabilities and asks for the highest expected payoff.

1. Check that each $P(S_j)$ is in $[0, 1]$ and that the probabilities sum to one.
2. For each row, multiply each payoff by its matching column probability and add the products, $\text{EMV}(A_i) = \sum_j P(S_j)P(A_i, S_j)$.
3. Compare the totals and report the highest expected payoff in the original units.

Fruit Packing

A cooperative chooses a packing plan before export demand is known. Payoffs and EMVs are in thousand USD.

Alternative	Strong	Steady	Weak
$P(S_j)$	0.40	0.35	0.25
Premium packs (A_1)	90	44	−12
Mixed packs (A_2)	68	58	24
Basic packs (A_3)	46	48	38

Each probability is in $[0, 1]$, and $0.40 + 0.35 + 0.25 = 1$.

$$\text{EMV}(A_1) = 0.40(90) + 0.35(44) + 0.25(-12) = 36 + 15.40 - 3 = 48.40,$$
$$\text{EMV}(A_2) = 0.40\,(\,68\,) + 0.35\,(58) + 0.25\,(24) = 27.20 + 20.30 + 6 = \mathbf{53.50},$$
$$\text{EMV}(A_3) = 0.40(46) + 0.35(48) + 0.25(38) = 18.40 + 16.80 + 9.50 = 44.70.$$

Decision. Choose **Mixed packs** (A_2). Under the stated probabilities, expected profit is **53.50** thousand USD (USD 53,500). The profit actually earned depends on which state occurs.

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C7 — Find where the EMV recommendation changes

When to use it. A state probability depends on p, and the question asks which option wins as p changes.

1. Use the sum-to-one condition to express the missing probability in terms of p.
2. Require every probability to lie in $[0, 1]$. Intersect the resulting bounds.
3. Write the competing EMVs as functions of p. Set them equal and check the tie is feasible.
4. Compare on each side and at the feasible endpoints. Report the winner by interval, including the tie and any infeasible values.

Online Course Launch

A training business chooses a launch plan before enrolment is known. Profits are in thousand USD.

Alternative	High (S_1)	Moderate (S_2)	Low (S_3)
$P(S_j)$	p	$p/2$	$1 - 3p/2$
Full (A_1)	100	36	−8
Pilot (A_2)	52	48	28

From $p + p/2 + P(S_3) = 1$, we get $P(S_3) = 1 - 3p/2$. The three probability checks give $0 \le p \le 1$, $0 \le p \le 2$, and $0 \le p \le 2/3$, respectively. The feasible range is their intersection, $0 \le p \le 2/3$.

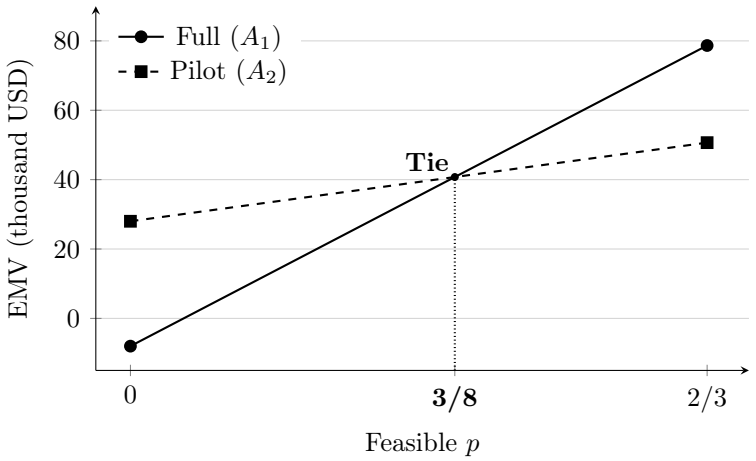
$$E_1(p) = 100p + 36(p/2) - 8(1 - 3p/2) = 130p - 8,$$
$$E_2(p) = 52p + 48(p/2) + 28(1 - 3p/2) = 28 + 34p.$$

The tie solves $130p - 8 = 28 + 34p$, so $96p = 36$ and $p = \mathbf{3/8}$ is feasible. Both EMVs equal 40.75 thousand USD there. The difference $E_1 - E_2 = 96p - 36$ is negative below the tie and positive above it. At $p = 0$, $(E_1, E_2) = (-8, 28)$. At $p = 2/3$, $(E_1, E_2) = (236/3, 152/3)$.

Decision. Choose the launch with the higher expected profit at your feasible value of p.

p	Recommendation
$0 \le p < \mathbf{3/8}$	Pilot (A_2)
$p = \mathbf{3/8}$	Both tied
$\mathbf{3/8} < p \le 2/3$	Full (A_1)

For $2/3 < p \le 1$, the low-enrolment probability is negative, so these inputs are infeasible.



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C8 — Compare rules and test payoff changes

When to use it. The question compares methods or asks when a revised payoff changes the recommendation.

1. Use the same starting payoff table for all four rules. Record each recommendation and its deciding value.
2. Change only the specified payoffs. Apply the stated eligibility rule before ranking the remaining alternatives.
3. Recalculate the relevant statistics on the eligible set. For regret, recompute the state benchmarks on that set.
4. Use a strict inequality for a sole winner. Check the tie and the first permitted integer beyond it.

Repair Workshop Plans

A bicycle repair workshop chooses its capacity. Profits and deciding values are in thousand USD.

	Payoffs			Decision statistics			
Alternative	High	Normal	Low	Max	Min	Max regret	EMV
$P(S_j)$	0.2	0.6	0.2				
Large (A_1)	120	25	−20	**120**	−20	55	35
Flexible (A_2)	85	65	−10	85	−10	**45**	54
Basic (A_3)	50	45	35	50	**35**	70	44
Partner (A_4)	70	75	20	75	20	50	**63**

The state bests are $(120, 75, 35)$, so A_2's maximum regret is $\max\{120-85, 75-65, 35-(-10)\} = \mathbf{45}$. The Partner plan's EMV is $0.2\,(70) + 0.6\,(75) + 0.2\,(20) = 14 + 45 + 4 = \mathbf{63}$.

Decision. Maximax chooses **Large** at **120**. Maximin chooses **Basic** at **35**. Minimax Regret chooses **Flexible** at **45**. EMV/Bayes chooses **Partner** at **63**.

Variant. Subsidy and Maximin. For integer $0 \le s \le 50$, replace the Low payoffs for A_1, A_2 with $-20+s, -10+s$. All payoffs must be nonnegative. A_1 becomes eligible at $s = 20$, A_2 at $s = 10$, and A_3, A_4 are always eligible. The row minima in order are $\min\{25, -20+s\}, -10+s, 35, 20$. Only A_2 can overtake A_3, since A_1's minimum is at most 25 and A_4's is 20. Flexible ties Basic at $s = \mathbf{45}$, since $-10 + 45 = 35$. It wins uniquely when $-10 + s > 35$. The first such integer is $s = \mathbf{46}$, when its minimum profit is 36 thousand USD.

Community Market Stall

Profits are in hundred USD. A sponsor replaces Compact's High payoff of 48 with an integer g in $[48, 80]$. Both payoffs must be at least 20.

Alternative	High	Low	Eligible for this range?
Large (C_1)	80	10	No
Compact (C_2)	g	32	Yes
Standard (C_3)	70	25	Yes
Premium (C_4)	90	−5	No

The original unrestricted benchmarks were $(90, 32)$, and Standard won on maximum regret 20. After screening, the benchmarks are $(\max\{g, 70\}, 32)$. Thus $R(C_2) = \max\{70-g, 0\}$ and $R(C_3) = \max\{g-70, 7\}$. Up to $g = 70$, compare $70-g$ with 7. At $g = \mathbf{63}$ both regrets equal 7. The first integer with $70-g < 7$ is $g = \mathbf{64}$. Compact then wins with regret 6 versus 7, and its high-attendance profit is USD 6,400.

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C9 — Apply criteria in stages

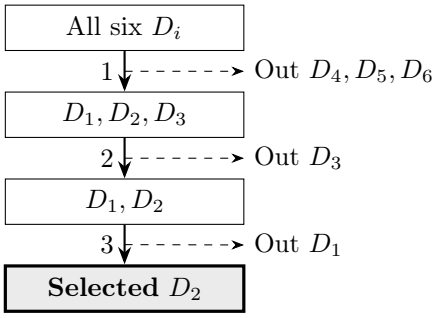
When to use it. The question gives an ordered list of rules and tells you how many alternatives to keep.

1. Start with all eligible alternatives. Apply the first rule and rank its deciding values in the correct direction.
2. Keep the required alternatives. Record survivors, eliminations, and their values. If a tie crosses a cutoff, use the problem's tie rule or report the unresolved tie if none is given.
3. At each later stage, compare only survivors. Use the supplied statistics when given. If the task calls for fresh regrets, find the benchmarks on the current comparison set.
4. Draw solid arrows for survivors and dashed arrows for eliminations. Mark the final choice or report a final tie under the stated rules.

Book Fair Stock

The bookseller compares plans D_1 through D_6. All deciding values below are in hundred USD.

Stage and rule	Survivors (value)	Eliminated (value)
0. Start	D_1 through D_6	—
1. EMV/Bayes, best 3	$D_1(65)$, $D_2(65)$, $D_3(65)$	$D_4(59)$, $D_5(53)$, $D_6(43)$
2. Maximax, best 2	$D_1(100)$, $D_2(90)$	$D_3(80)$
3. Maximin, final	$D_2(\mathbf{10})$	$D_1(0)$



All three plans with EMV 65 fit into the three available places. Next $100 > 90 > 80$ leaves D_1, D_2.

Decision. The bookseller selects D_2. Of the two finalists, it has the higher worst-case profit, **10** hundred USD (USD 1,000).

C10 — Justify a criterion in context

When to use it. The question asks which rule suits the decision maker and why.

1. Check whether credible state probabilities are available.
2. Identify the decision maker's priority and willingness to accept a poor outcome.
3. Choose the rule that fits, show its deciding calculation, and explain what the result means for this decision.

Sportswear Stock, with a cash constraint

Suppose the retailer has no reliable demand probabilities and needs to protect cash in a weak season. The row minima from **C4** are $-30, 16, 29, 24$ thousand USD. Maximin fits that priority because it compares the worst outcomes. Since $\max\{-30, 16, 29, 24\} = \mathbf{29}$, the retailer should choose **Small order**, whose lowest listed profit is **29** thousand USD.

Interactive Decision Webpage activity checklist. Use the same reasoning to interpret the webpage's recommendations. Check that users can do the following.

- ☐ Add, edit, and remove alternatives and states. Edit payoffs and state probabilities.
- ☐ Compare all four rules with their values, recommendations, and any ties.
- ☐ Apply staged filtering and see each elimination and the final result in a flow diagram.
- ☐ Select at least two distinct preloaded scenarios, each with alternatives, states, payoffs, and probabilities, to test all four rules, filtering, and the diagram.
- ☐ See calculations and visuals update automatically. Check that the layout is clear and responsive, results agree, and invalid inputs are handled.
- ☐ Find a visible, accessible link to the AI development conversation.

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